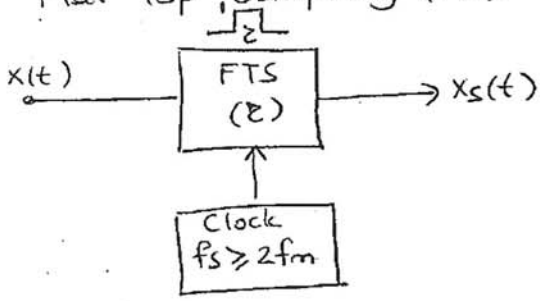
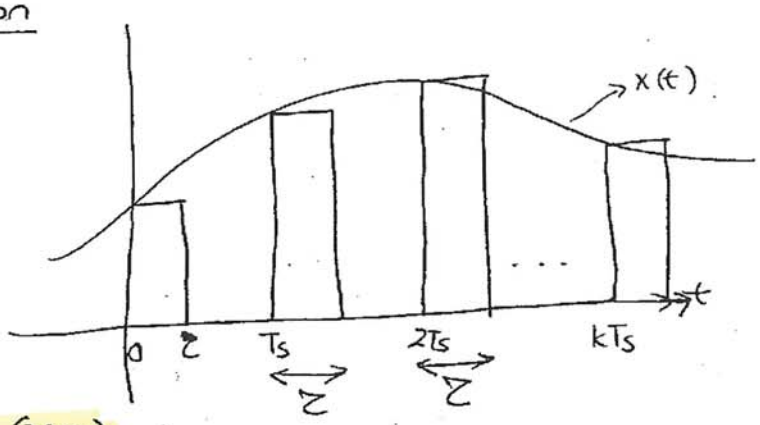
Sample and Hold Operation (Örnekle ve tut işlemi)

Flat Top Sampling (FTS)
(Düz Tepeli Örnekleme)

Sample and Hold Operation

Flat Top Sampling (FTS)

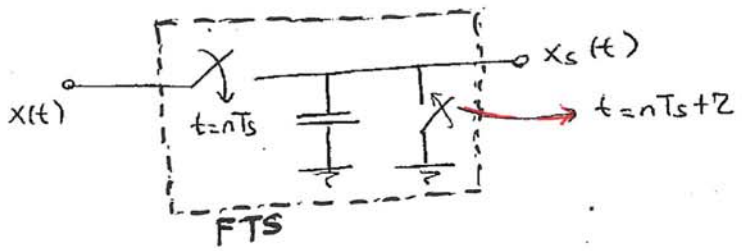


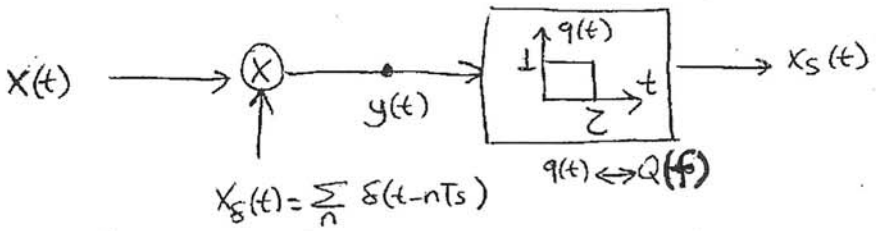


xs(t): Pulse Amplitude Modulation (PAM)

Discrete time analog modulation

Sample / Hold (Dump) Circuit (Örnekle ve tut devresi)





$$y(t) = x(t)\,x_\delta(t) = \sum_n x(nT_s)\,\delta(t-nT_s)$$

$$x_s(t) = q(t) \circledast y(t) = \sum_n x(nT_s)\,q(t-nT_s)$$

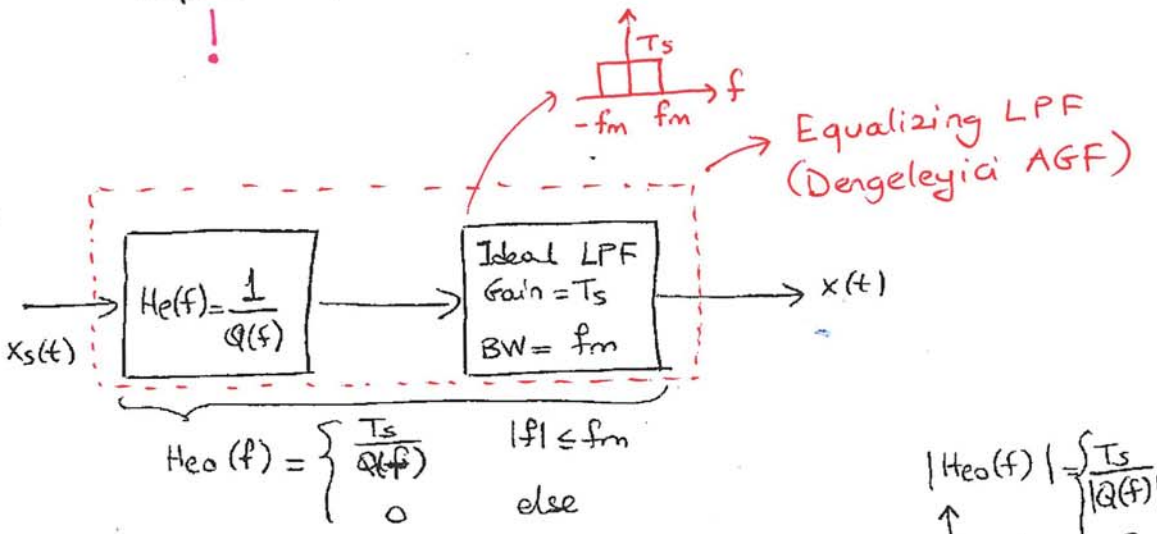
$$x_s(t) = q(t) \circledast \left[x(t)\,x_\delta(t)\right] = q(t) \circledast \left[\sum_n x(nT_s)\,\delta(t-nT_s)\right] = \sum_n x(nT_s)\,q(t-nT_s)$$

$$X_s(f) = Q(f).\left[X(f) \circledast f_s \sum_n \delta(f-nf_s)\right]$$

$$X_s(f) = Q(f).\left[f_s \sum_n X(f-nf_s)\right]$$

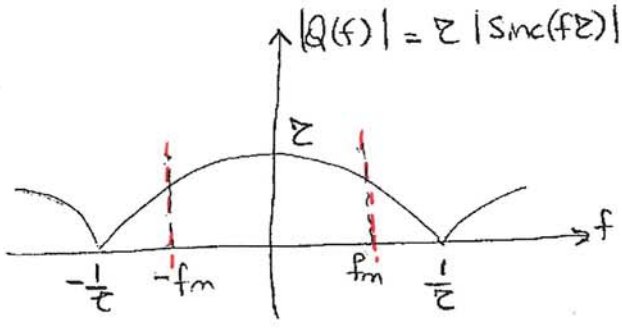
$$\hookrightarrow \because \; \tau\,\mathrm{sinc}(f\tau).\,e^{-j2\pi f\frac{\tau}{2}}$$

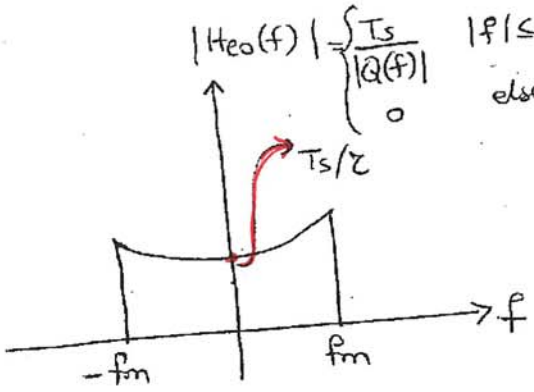
$$X_s(f) = \underbrace{f_s\,Q(f)X(f)}_{!} + Q(f).f_s \sum_{n\neq 0} X(f-nf_s)$$



$$H_{eo}(f) = \begin{cases} \dfrac{T_s}{Q(f)} & |f| \le f_m \\ 0 & \text{else} \end{cases}$$

$$|H_{eo}(f)| = \begin{cases} \dfrac{T_s}{|Q(f)|} & |f| \le f_m \\ 0 & \text{else} \end{cases}$$





$$H_{eo}(f) = \frac{T_s}{Q(f)} = \frac{T_s}{\tau\,\mathrm{Sinc}(f\tau)}\,e^{j2\pi f\frac{\tau}{2}}, \quad |f| \le f_m$$

↳ advance in time → heo(t) is not causal

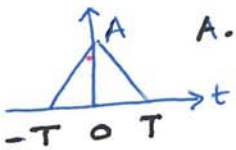
For this reason, $H'_{eo}(f) = H_{eo}(f).\,e^{-j2\pi f T_x}$ is chosen

$$\downarrow$$

$$h'_{eo}(t) = h_{eo}(t-T_x) \qquad T_x = \frac{\tau}{2} + T_0$$

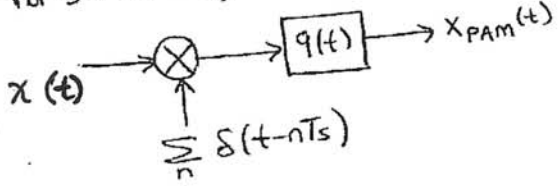
$$x_s(t) \longrightarrow \boxed{H'_{eo}(f)} \longrightarrow x(t-T_0)$$

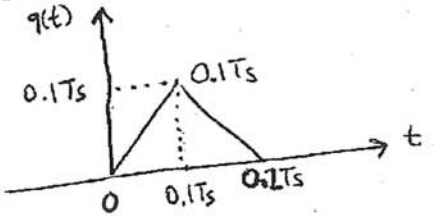
Reminder:



$$A \cdot \Lambda\left(\frac{t}{T}\right) \overset{\mathcal{F}}{\longleftrightarrow} AT\,\mathrm{Sinc}^2(fT)$$

Ex. (2004 Midterm Ex. Q.)
A PAM signal is produced as shown in the figure below. a) If the bandwidth of $x(t)$ is 1 kHz, what is the maximum value of T_s to obtain $x(t)$ without distortion at the receiver?
b) For $T_s = T_s(max)$, sketch PAM demodulator. Find the transfer function of demodulator.





a) $BW\{x(t)\} = 1\ kHz = f_m$

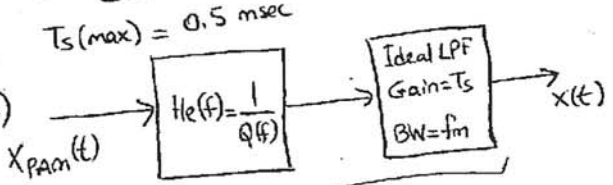
$f_s \geq 2f_m = 2\ kHz$

$T_s \leq \frac{1}{2f_m} = 0.5\ msec$

$T_s(max) = 0.5\ msec$

$$Q(f) = 0.1T_s \times 0.1T_s\,\mathrm{Sinc}^2(0.1T_s f)\, e^{-j2\pi f 0.1T_s}$$
$$= 0.01\,T_s^2\,\mathrm{Sinc}^2(0.1T_s f)\, e^{-j2\pi f 0.1T_s}$$

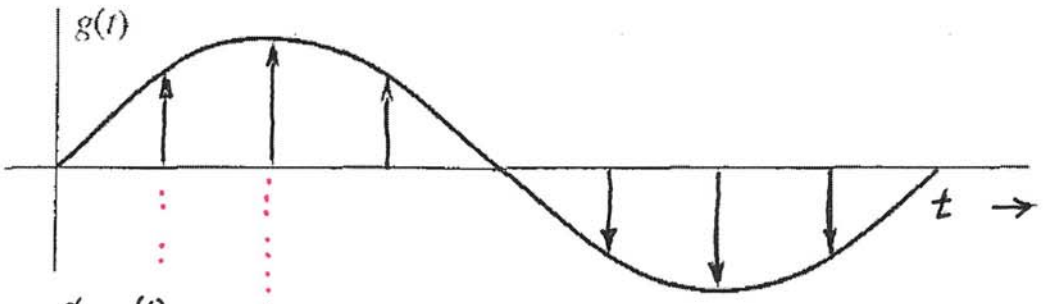
b)

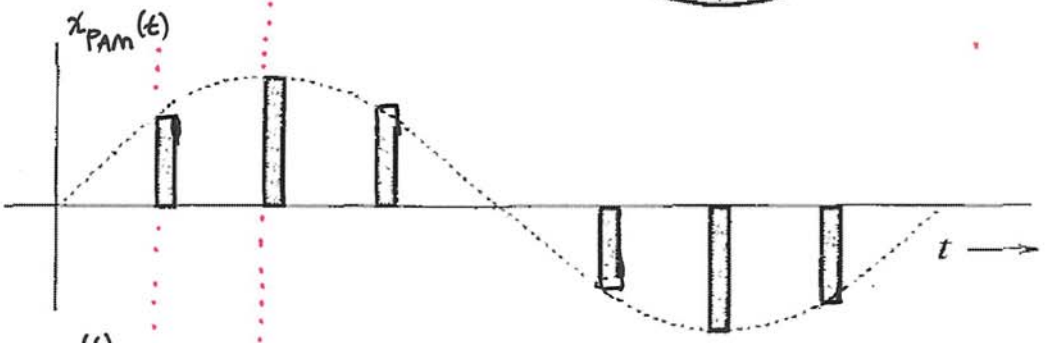


$$H_{eo}(f) = \begin{cases} \frac{T_s}{Q(f)}, & |f| \leq f_m \\ 0, & |f| > f_m \end{cases}$$

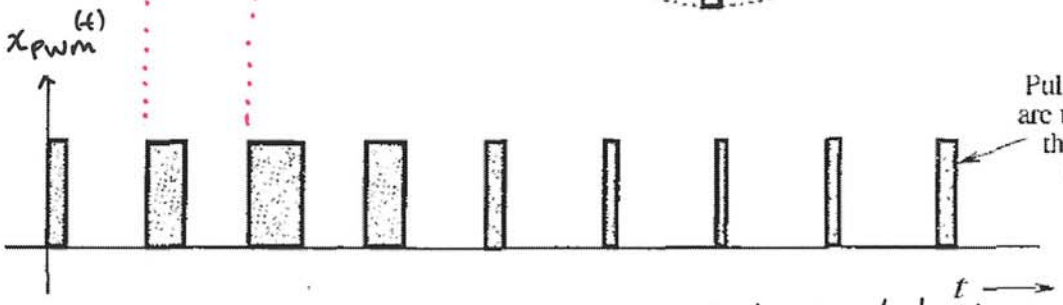
$$H_{eo}(f) = \frac{T_s \cdot e^{+j2\pi f 0.1T_s}}{0.01\,T_s^2\,\mathrm{Sinc}^2(0.1T_s f)}$$

$$= \begin{cases} \frac{2\times10^5\, e^{j2\pi f 5\times10^{-5}}}{\mathrm{Sinc}^2(5\times10^{-5} f)}, & |f| \leq 1\ kHz \\ 0, & \text{else} \end{cases}$$



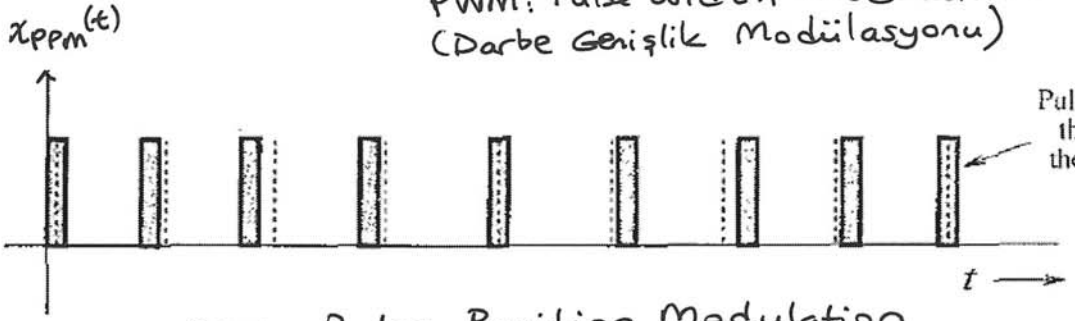


PAM



PWM

PWM: Pulse Width Modulation
(Darbe Genişlik Modülasyonu)

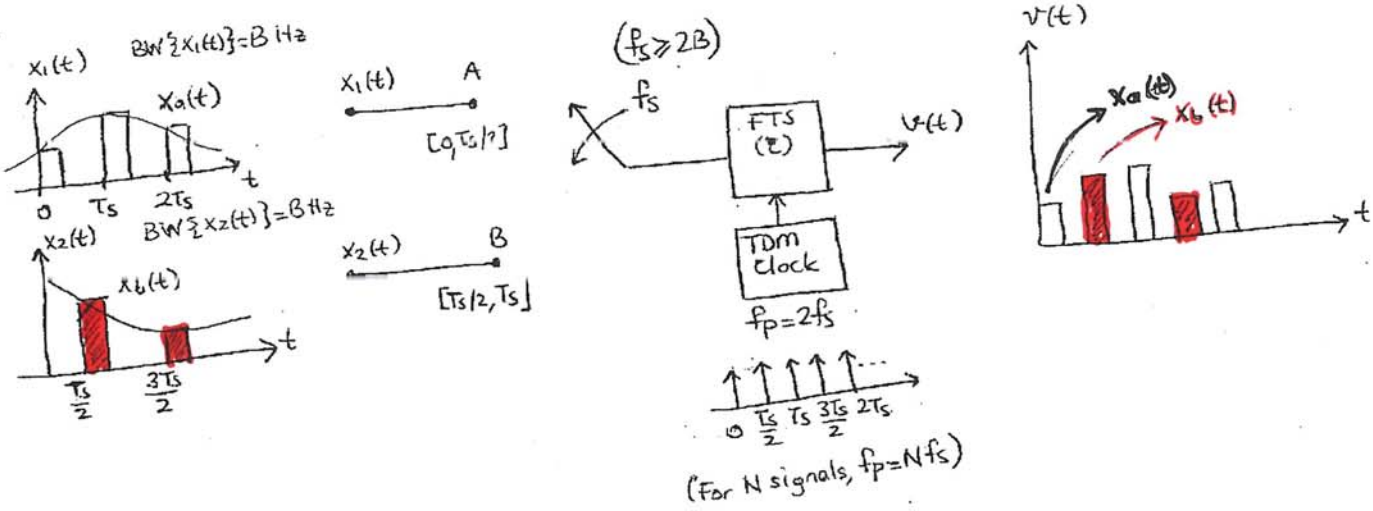


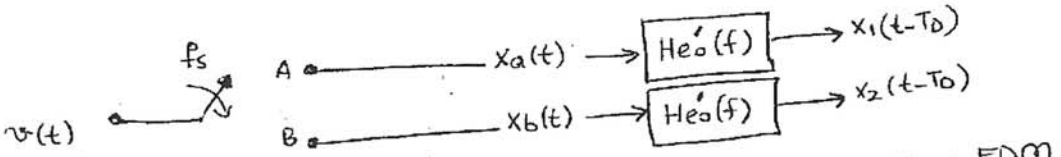
PPM

PPM: Pulse Position Modulation
(Darbe Pozisyon Modülasyonu)

Time Division Multiplexing (TDM) — Zaman Bölmeli Çoklama (ZBÇ)

- Multiple PAM signals are transmitted without overlapping.
- At the receiver, PAM signals are separated by synchronization.





FDM x TDM

Advantages: 1) Easier to implement. 2) Less crosstalk than FDM

Disadvantage: 1) strict synchronization

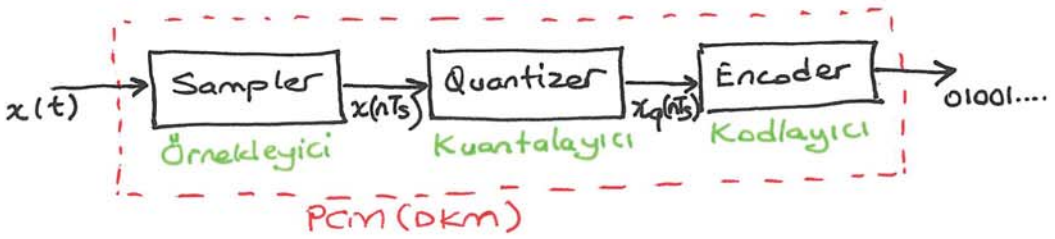
Çok kaynaklı (Multi-source) ⟶ Multiplexing (Çoklama)

Çok kullanıcılı (Multi-user) ⟶ Multiple Access (Çoklu Erişim)

⟶ TDMA, FDMA

Pulse Code Modulation (PCM)

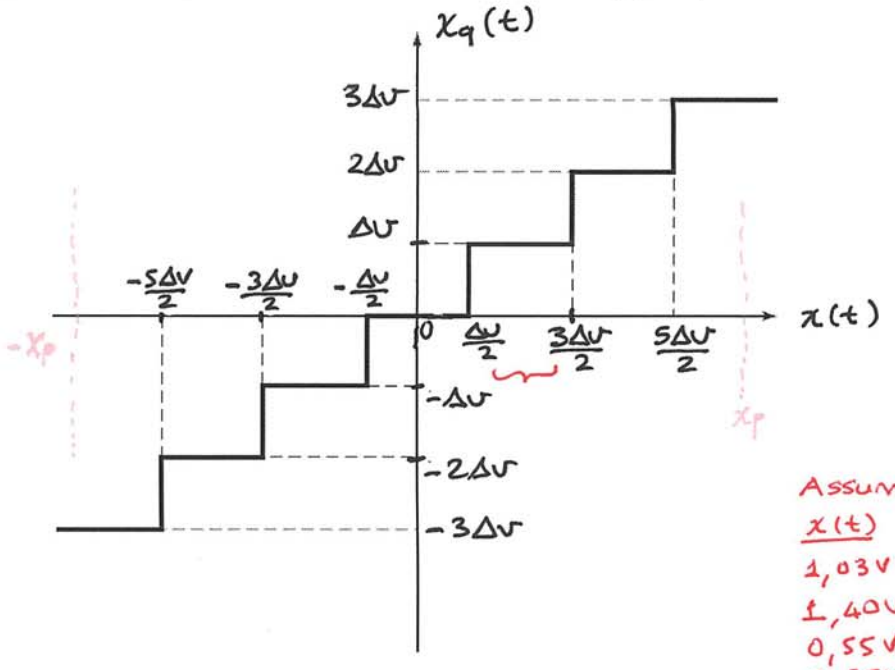
(Darbe Kod Modülasyonu - DKM)



$x(t)$ analog signal

$x_p = \text{maks}|x(t)|$ Peak value (Tepe değer)

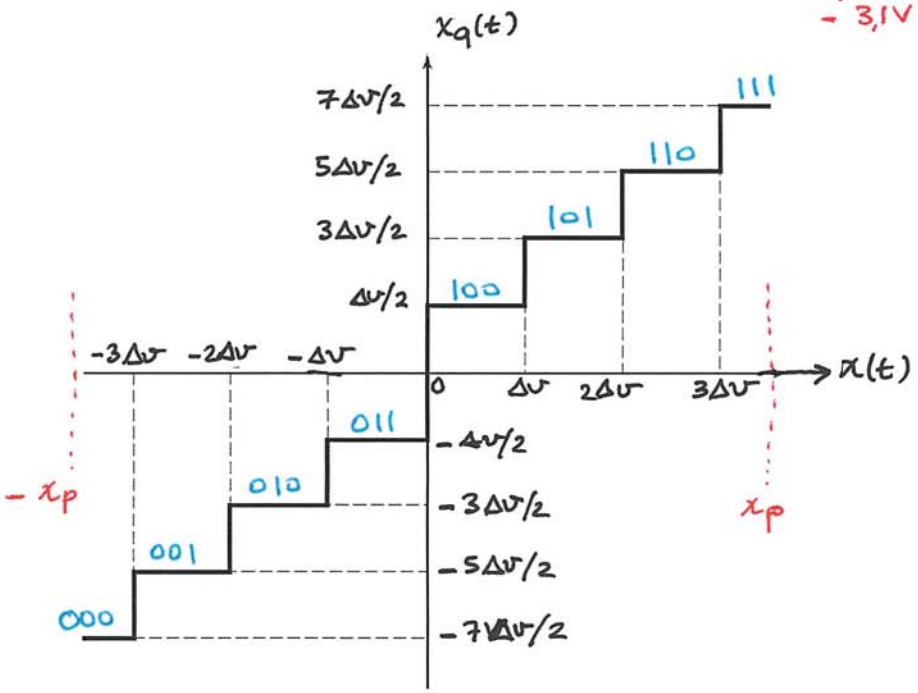
Kuantalayıcı adımı (Quantizer step) $= \Delta v = \dfrac{2x_p}{L}$ → Kuantalayıcadaki seviye sayısı (# of levels in quantizer)



$L = 7$

Assume $\Delta v = 1$ V

$x(t)$	$x_q(t)$
1,03 V	→ 1 V
1,40 V	→ 1 V
0,55 V	→ 1 V
2,25 V	→ 2 V
-3,1 V	→ -3 V



$L = 8$
(mid-rise quantizer)

Assume $\Delta v = 1$ V

$x(t)$	$x_q(t)$
1,03 V	1,5 V
0,55 V	0,5 V
-3,1 V	-3,5 V

* Each sample is approximated with a quantized value, the approximation will result in an error less than $\Delta v/2$ in the positive direction or $-\Delta v/2$ in the negative "

∴ Quantization error range: $(-\Delta v/2, \Delta v/2)$

$x_q(t) = x(t) + q(t)$ → Kuantalama gürültüsü (Quantization noise)

↳ Analog işaret

↳ Kuantalanmış işaret

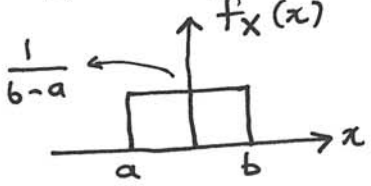
q is uniform in $[-\Delta v/2, \Delta v/2]$

Average quantization noise power = $N_q = E[q^2]$

(Ort. kuantalama gürültüsü gücü)

Reminder

X RD $[a,b]$ düzgün d.



$\mu_X = \frac{a+b}{2}$

$\sigma_X^2 = \frac{(b-a)^2}{12}$

$\sigma_q^2 = E[(q-\mu_q)^2] = E[q^2] = N_q$ ($\mu_q = 0$)

$N_q = \frac{[\Delta v/2 - (-\Delta v/2)]^2}{12} = \frac{\Delta v^2}{12}$

$\Delta v = \frac{2x_p}{L}$ → $N_q = \frac{\Delta v^2}{12} = \frac{x_p^2}{3L^2}$

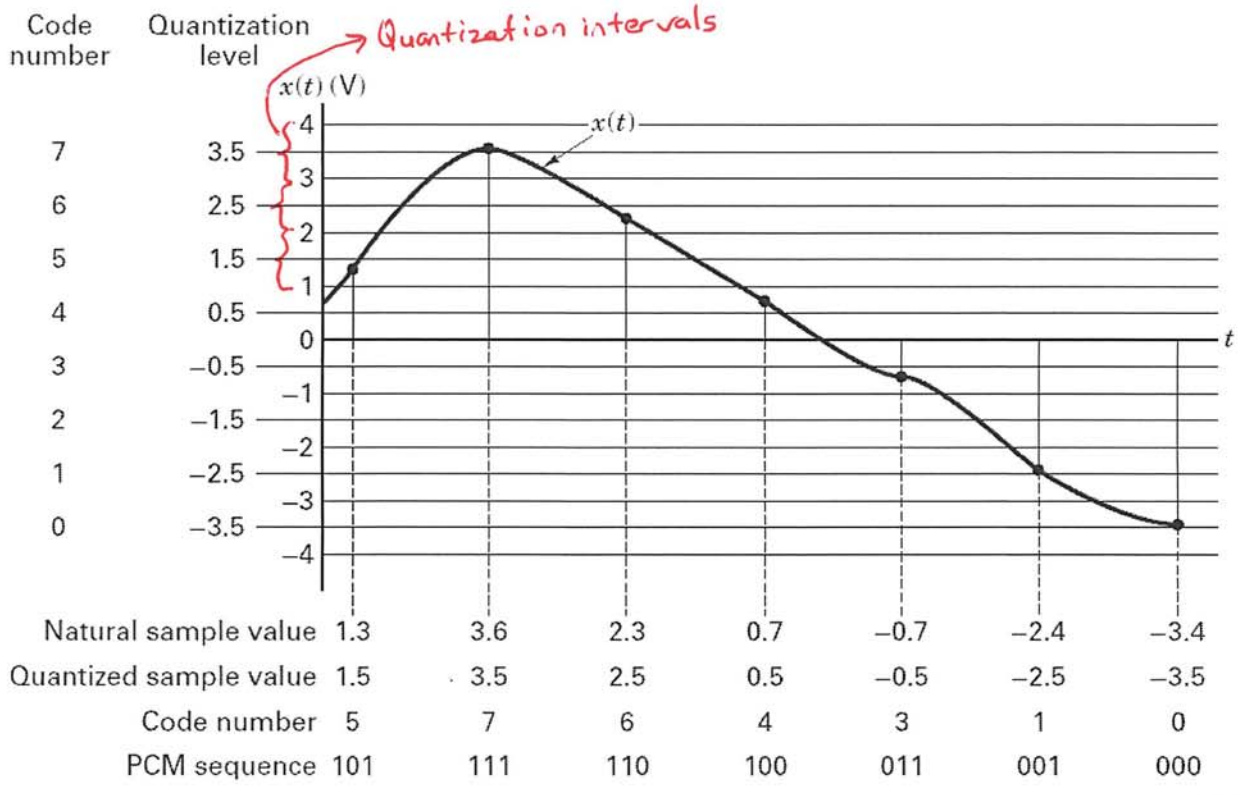
İşaretin gücü = $S_o = \langle x^2(t) \rangle = P_x$

Gürültünün gücü = $N_o = N_q$

İşaretin Gürültüye Oranı (İGO) } Signal-to-Noise Ratio (SNR)

$$SNR = \frac{S_o}{N_o} = \frac{P_x}{x_p^2} \cdot 3L^2$$

→ Signal-to-Quantization Noise Ratio (SQNR)



* Each sample is quantized to the middle of the quantization interval.